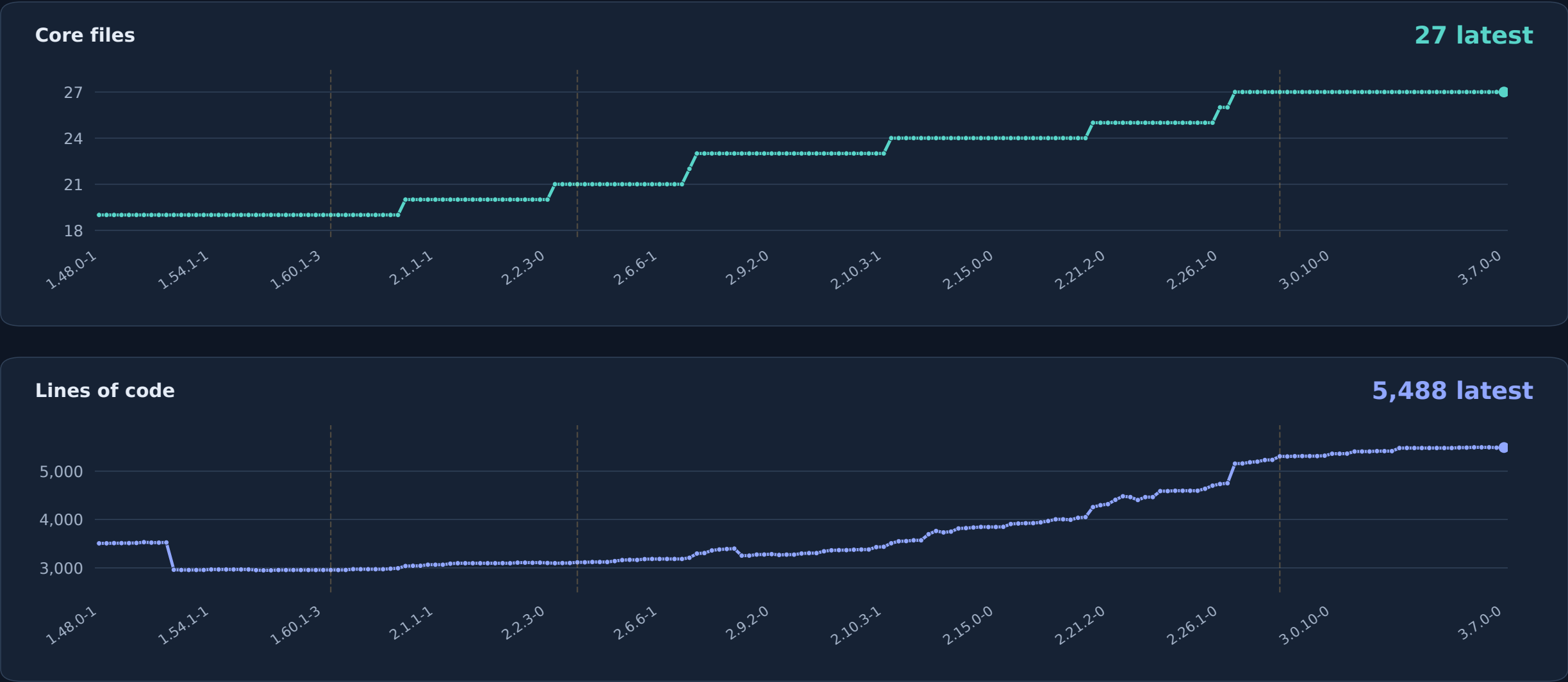


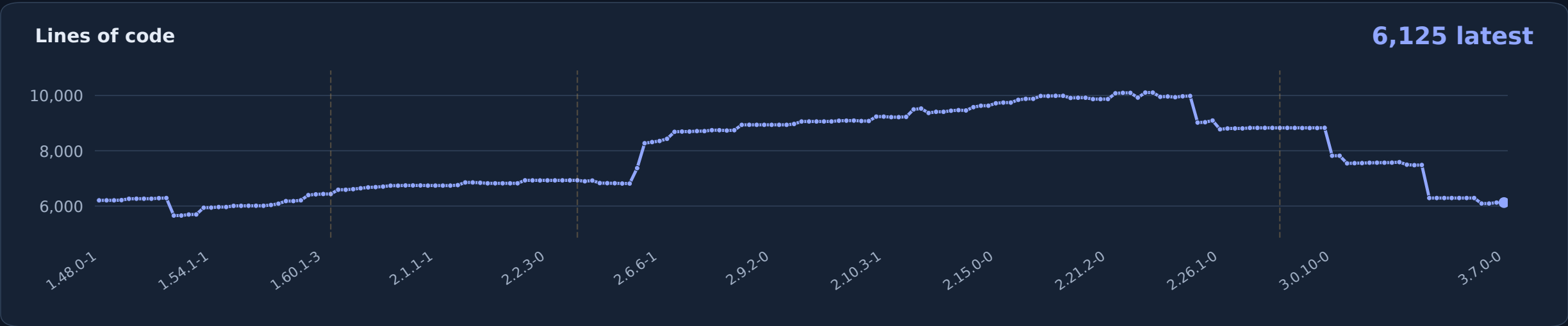
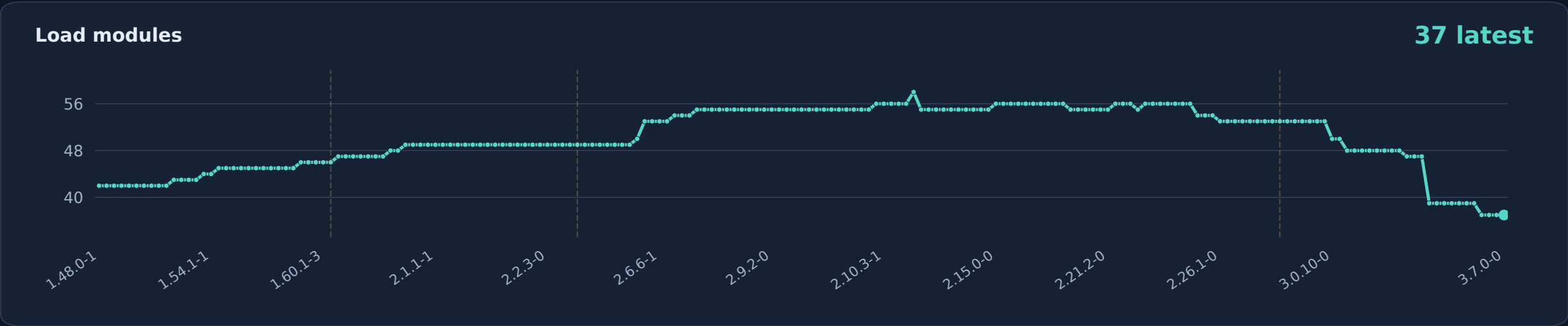
Core system evolution

1.48.0-1 to 3.7.0-0 / 189 recorded versions



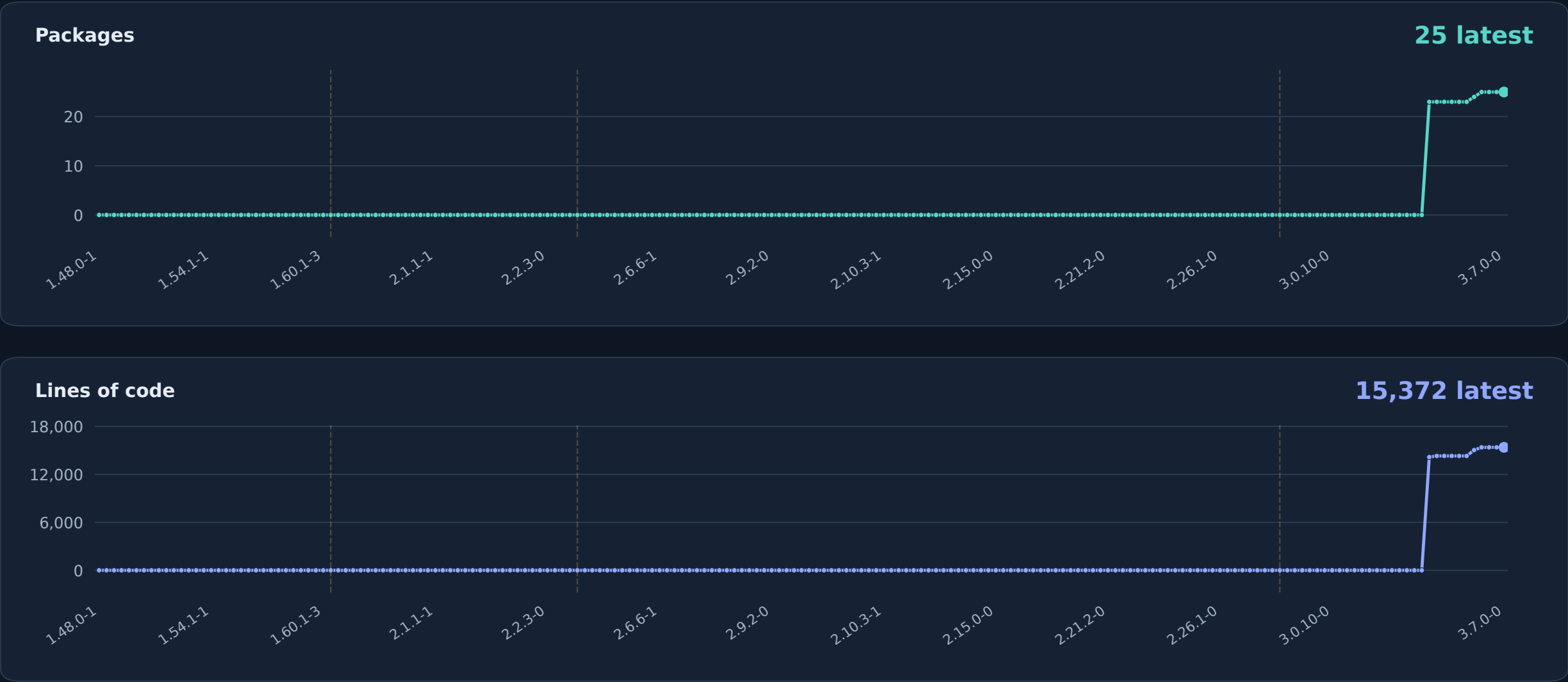
Load module evolution

1.48.0-1 to 3.7.0-0 / 189 recorded versions



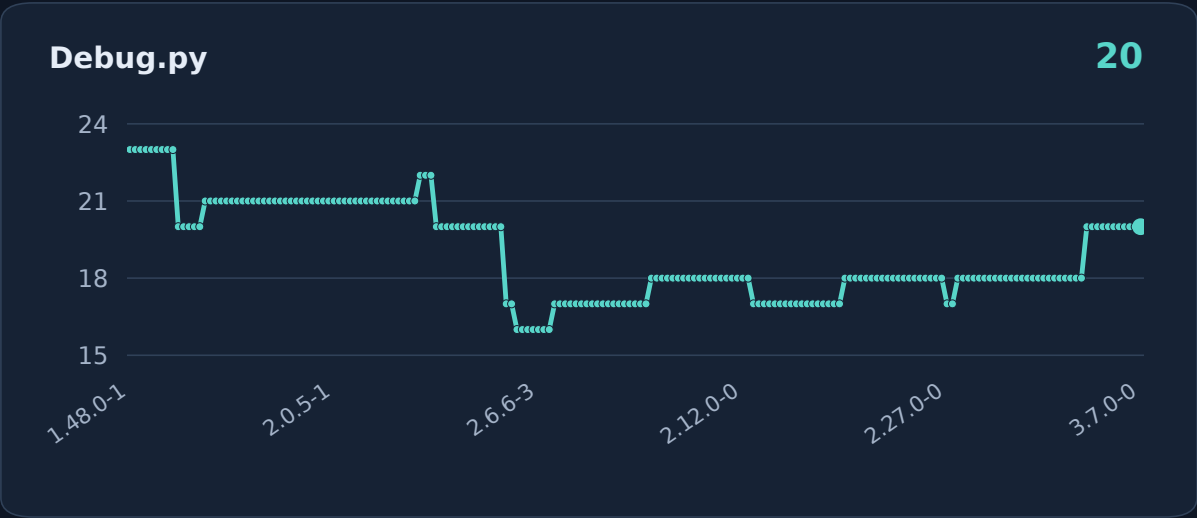
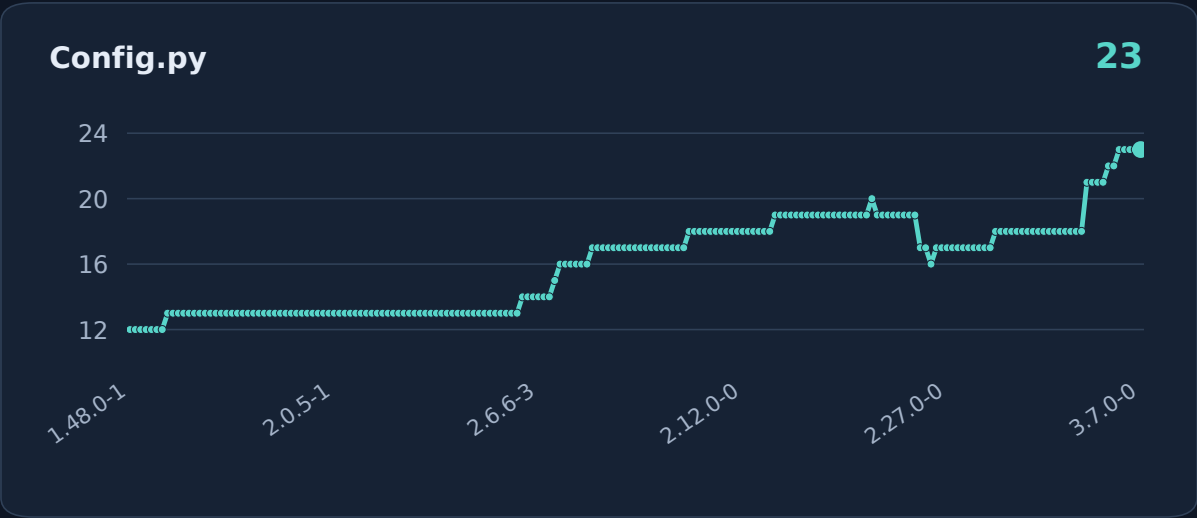
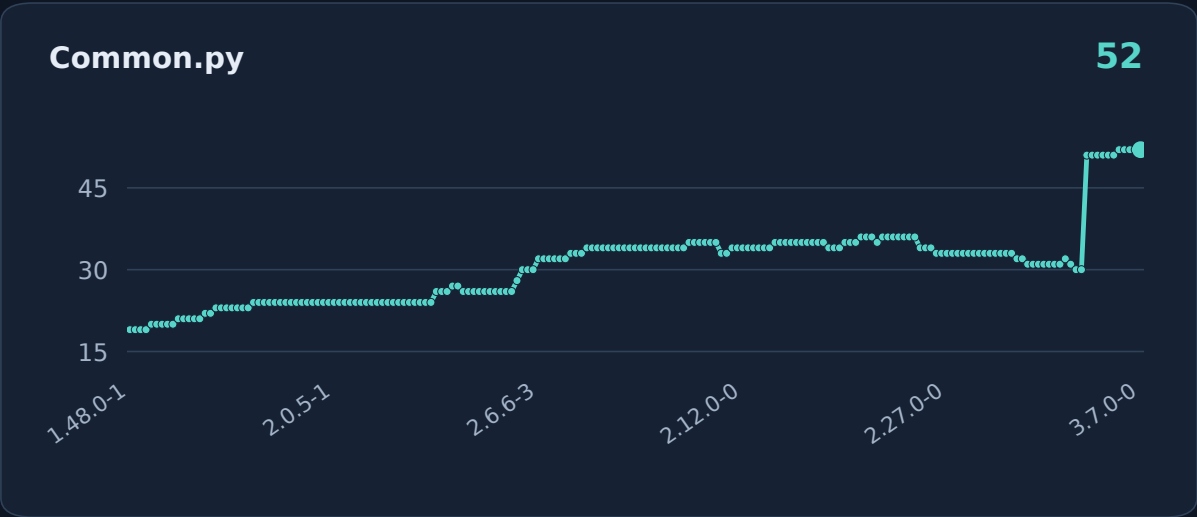
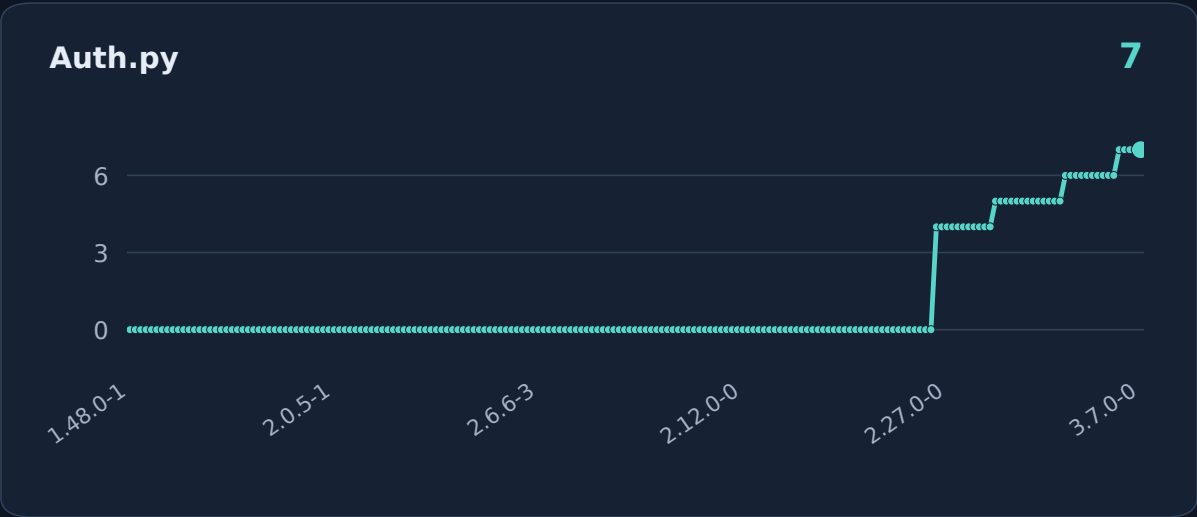
Package evolution

1.48.0-1 to 3.7.0-0 / 189 recorded versions



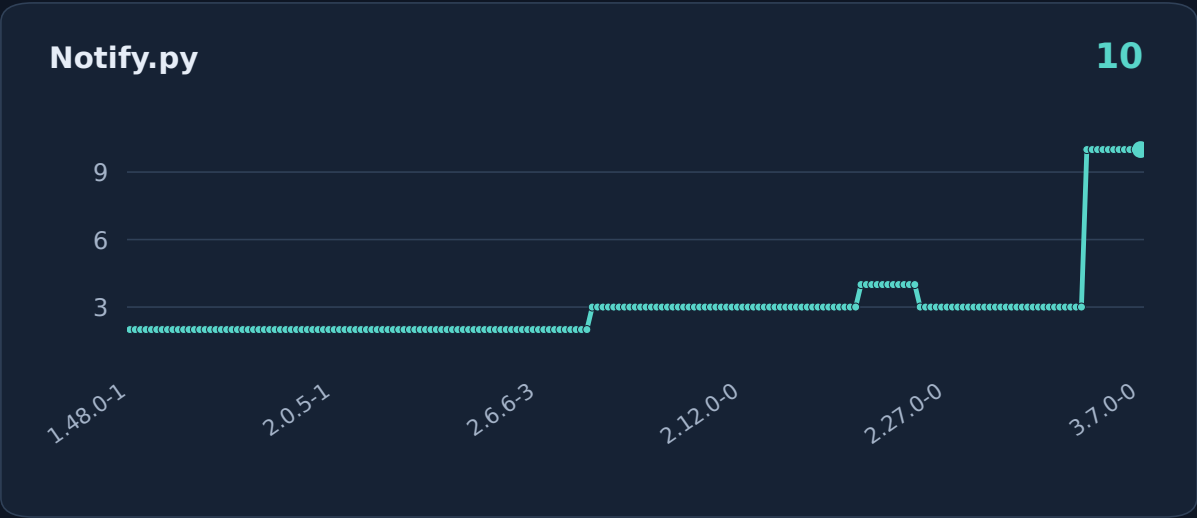
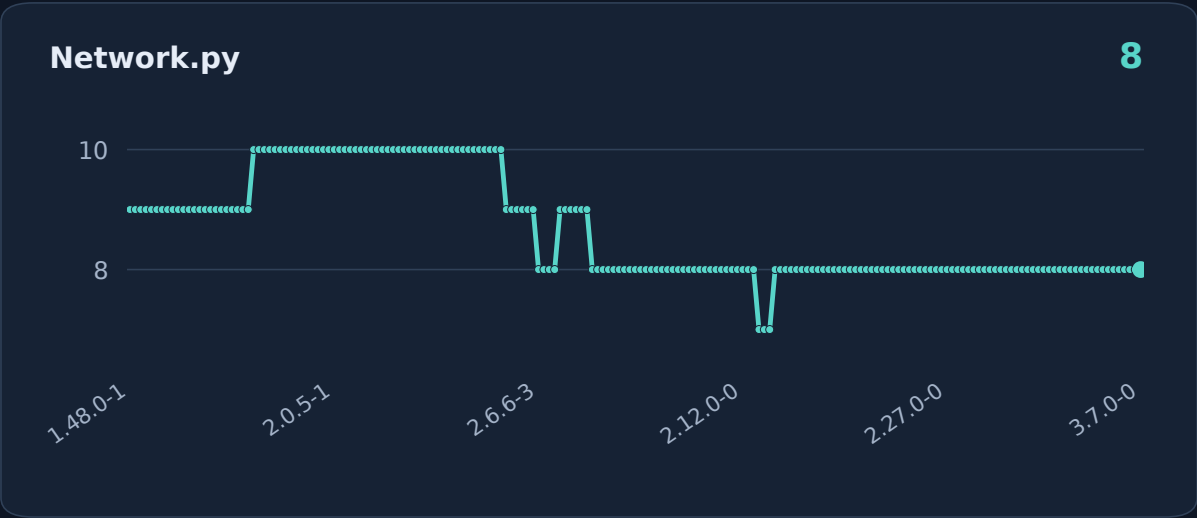
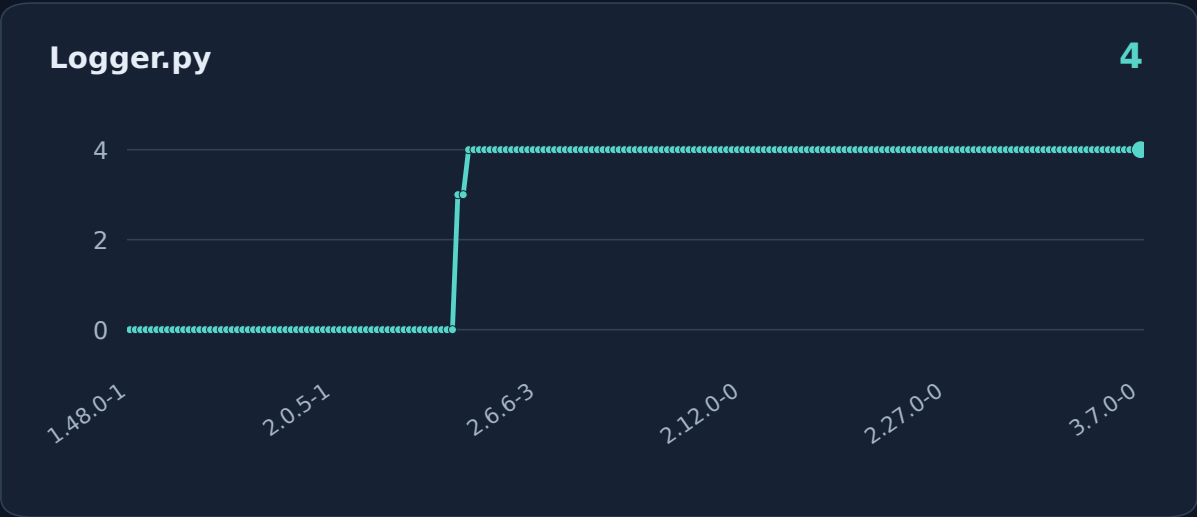
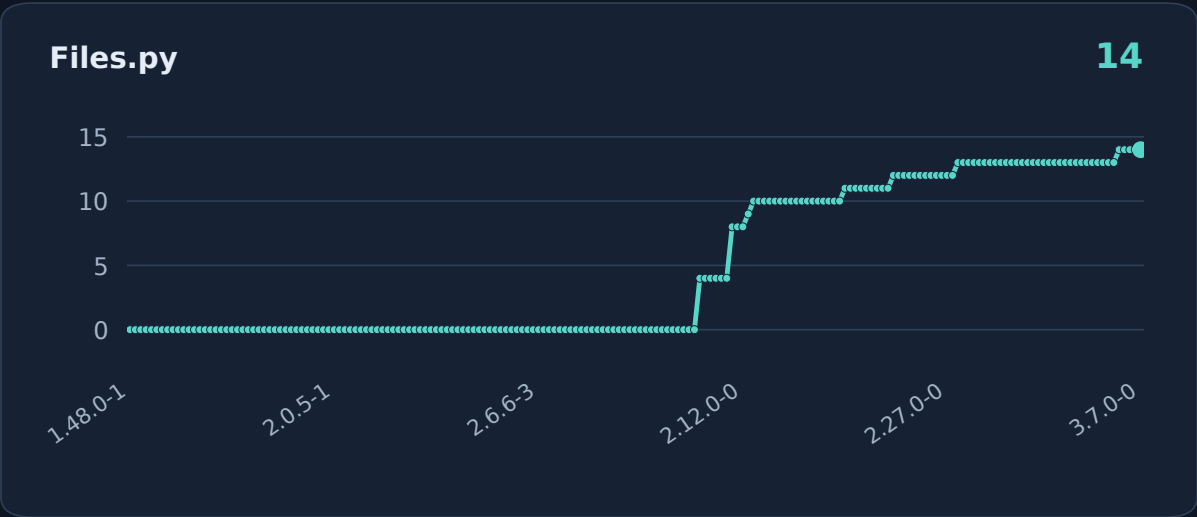
Core references

Per-file histories / latest reference count above 3



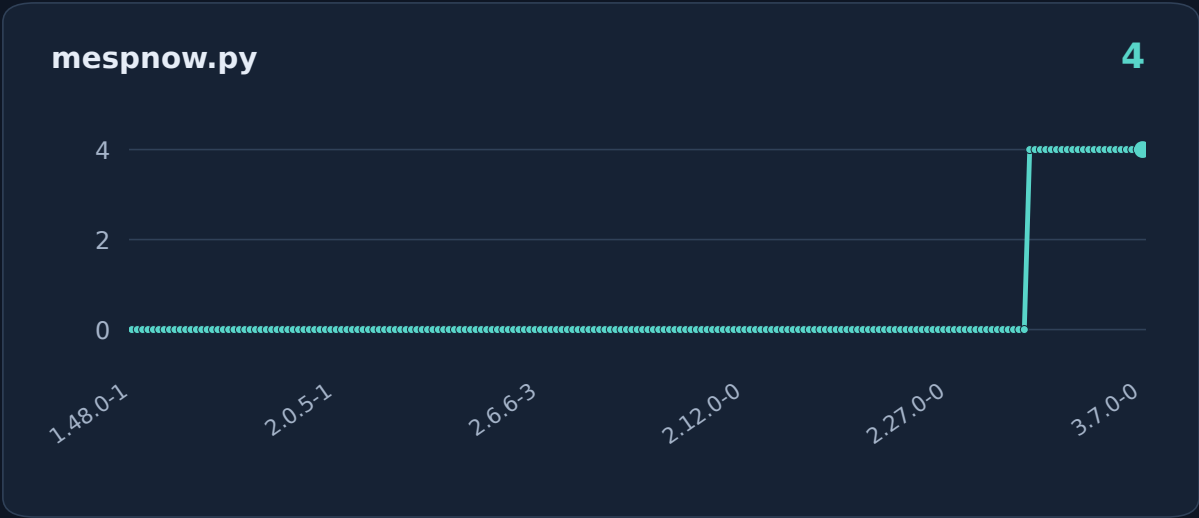
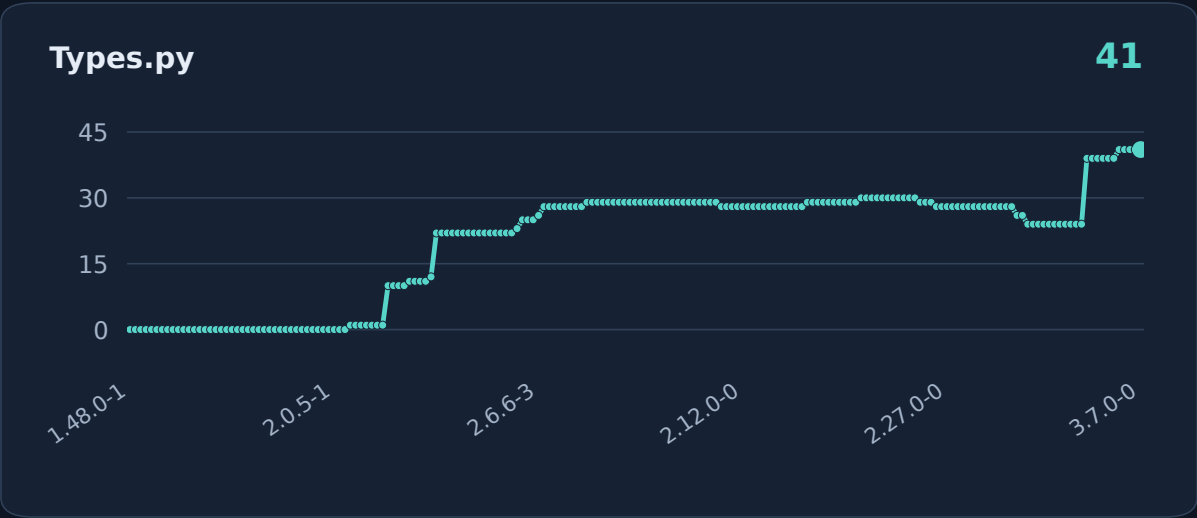
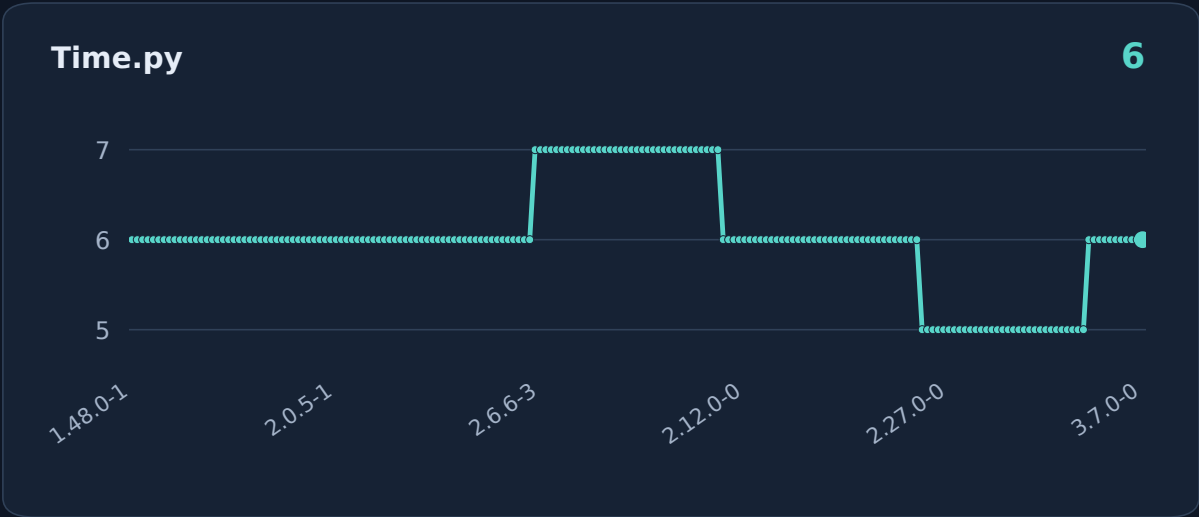
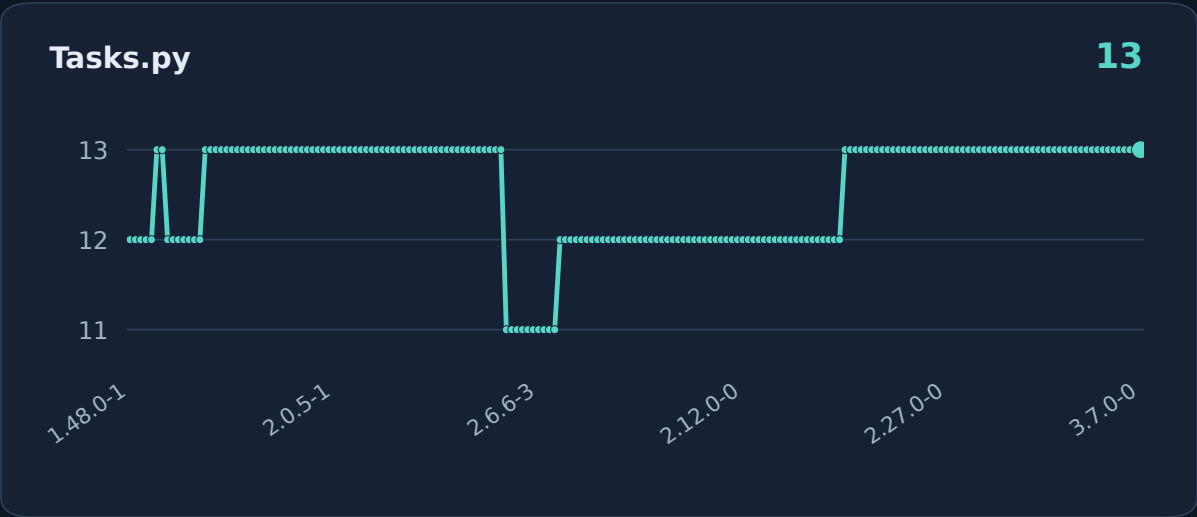
Core references

Per-file histories / latest reference count above 3



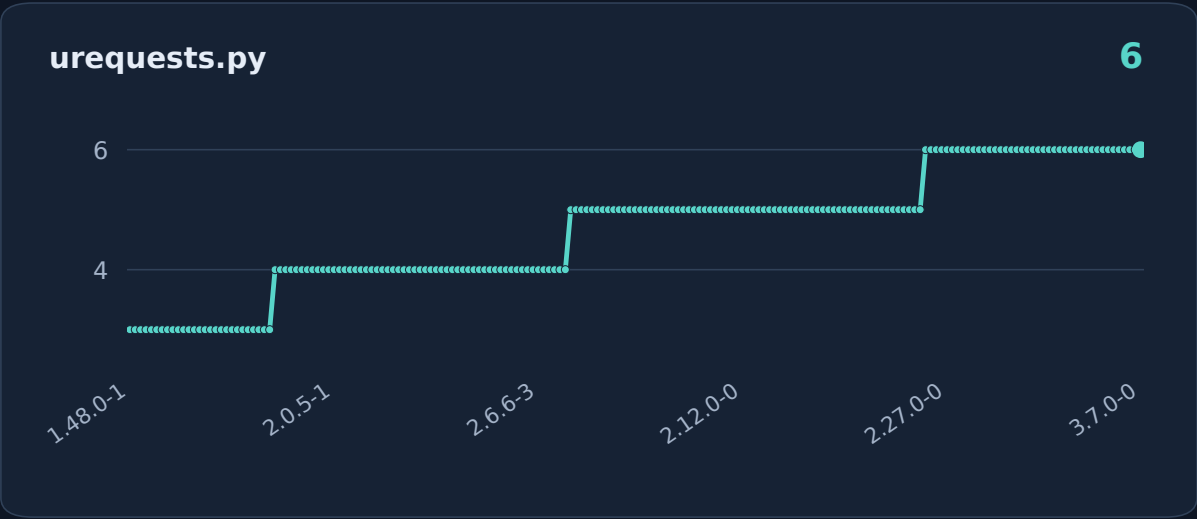
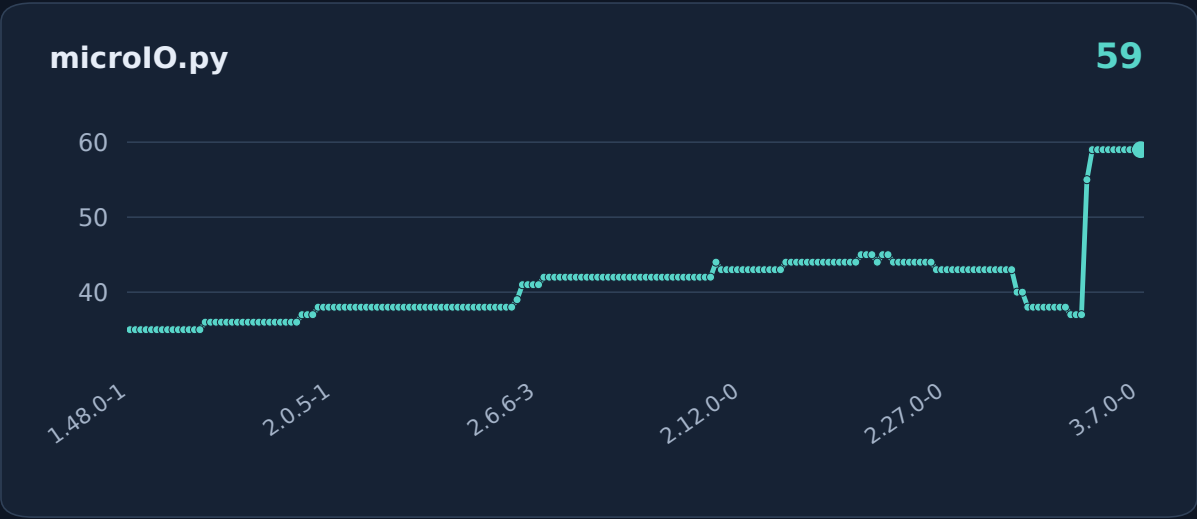
Core references

Per-file histories / latest reference count above 3



Core references

Per-file histories / latest reference count above 3



Lower-reference core files

Latest count of 3 or fewer / excluded from the trend charts

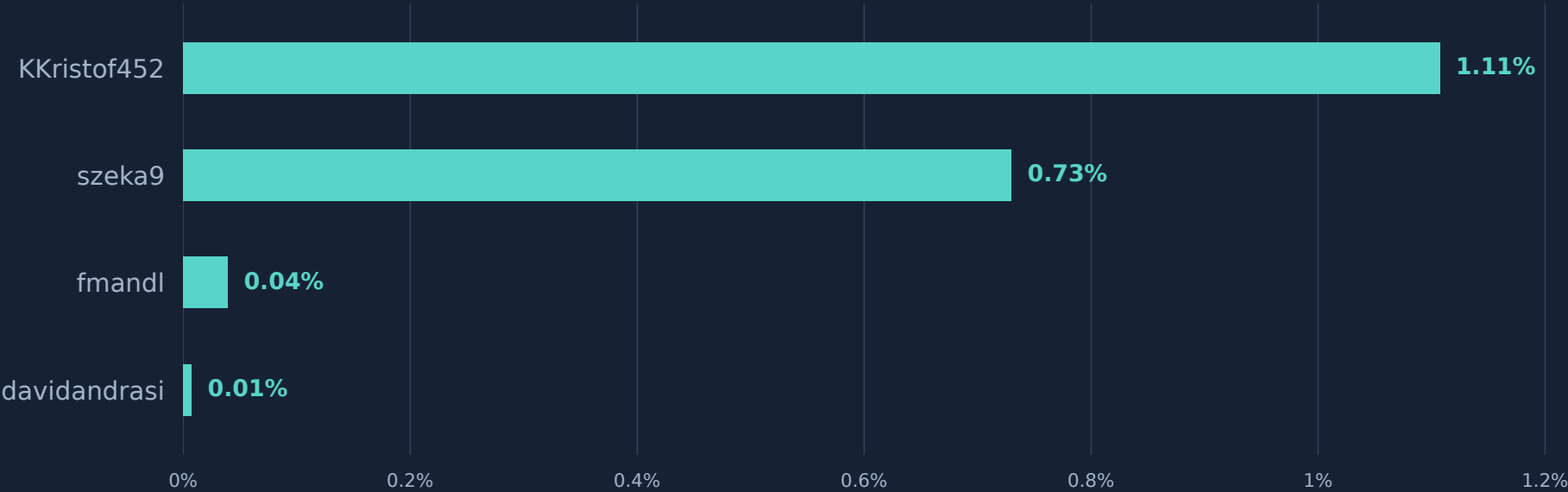
File	Latest references
Buffer.py	1
Espnow.py	0
Hooks.py	1
InterConnect.py	2
Interrupts.py	1
Pacman.py	3
Scheduler.py	1
Server.py	3
Shell.py	2
Web.py	2
helpers.py	0
main.py	0
micrOS.py	1
micrOSloader.py	1
reset.py	0

Project contributors

Share of project contributions / lead contributors shown above

BNM

98.12%



The bar chart shows contributors with a share of 30% or less.

Contributors' file changes

Complete recorded file lists / long paths wrap and continue onto the next card

KKristof452 / 56 file entries

apps/SysCheck_app.py
micrOS/ConfigHandler.py
micrOS/LM_L298N_DCmotor.py
micrOS/LM_system.py
micrOS/LM_system.py
micrOS/Network.py
micrOS/Scheduler.py
micrOS/SocketServer.py
micrOS/Time.py
micrOS/Time.py
micrOS/source/Espnow.py
micrOS/source/Espnow.py
micrOS/source/InterConnect.py
micrOS/source/InterConnect.py
micrOS/source/InterpreterShell.py
micrOS/source/LM_ds18.py
micrOS/source/LM_keychain.py
micrOS/source/LM_neoeffects.py
micrOS/source/LM_neopixel.py
micrOS/source/LM_oled_ui.py
micrOS/source/LM_rgb.py
micrOS/source/LM_system.py
micrOS/source/Scheduler.py
micrOS/source/Shell.py
micrOS/source/Shell.py
micrOS/source/SocketServer.py
micrOS/source/Tasks.py

KKristof452 / 56 file entries / continued

micrOS/source/Tasks.py
micrOS/source/Time.py
micrOS/source/Web.py
micrOS/source/dashboard.html
micrOS/source/micrOS.py
micrOS/source/modules/LM_espnw.py
micrOS/source/modules/LM_oled_ui.py
micrOS/source/modules/LM_oledui.py
micrOS/source/modules/LM_web.py
micrOS/source/udashboard.js
micrOS/source/urequests.py
micrOS/source/ustyle.css
micrOS/source/uwidgets.js
micrOS/source/web/config.css
micrOS/source/web/config.html
micrOS/source/web/config.js
micrOS/source/web/index.html
micrOS/source/web/uapi.js
micrOS/source/web/uapi_widgets.js
micrOS/source/web/ustyle.css
micrOS/utests/test_web_engine.py
toolkit/DevEnvCompile.py
toolkit/dashboard_apps/uLightDemo.py
toolkit/workspace/precompiled/modules/LM_espnw.py
toolkit/workspace/precompiled/web/config.css
toolkit/workspace/precompiled/web/config.html
toolkit/workspace/precompiled/web/config.js

Contributors' file changes

Complete recorded file lists / long paths wrap and continue onto the next card

KKristof452 / 56 file entries / continued

toolkit/workspace/precompiled/web/uapi.js
toolkit/workspace/precompiled/web/ustyle.css

szeka9 / 36 file entries

APPLICATION_GUIDE.md
README.md
env/requirements.txt
media/lms/i2s_mic.png
micrOS/source/Buffer.py
micrOS/source/Common.py
micrOS/source/Config.py
micrOS/source/LM_OV2640.py
micrOS/source/LM_aht10.py
micrOS/source/LM_i2s_mic.py
micrOS/source/LM_presence.py
micrOS/source/LM_sound_event.py
micrOS/source/Server.py
micrOS/source/Shell.py
micrOS/source/Tasks.py
micrOS/source/Web.py
micrOS/source/modules/LM_OV2640.py
micrOS/source/modules/LM_fileserver.py
micrOS/source/modules/LM_i2s_mic.py
micrOS/source/web/editor.js
micrOS/source/web/filesui.html
micrOS/source/web/filesui.js
micrOS/utests/test_buffer.py
micrOS/utests/test_web_engine.py
setup.py
toolkit/LM_to_compile.dat
toolkit/dashboard_apps/MicrophoneTest_app.py

Contributors' file changes

Complete recorded file lists / long paths wrap and continue onto the next card

szeka9 / 36 file entries / continued

```
toolkit/dashboard_apps/PresenceTest.py
toolkit/dashboard_apps/_capture.py
toolkit/dashboard_apps/_micPlotting.py
toolkit/dashboard_apps/_presencePlotting.py
toolkit/simulator_lib/machine.py
toolkit/workspace/precompiled/LM_presence.py
toolkit/workspace/precompiled/web/editor.js
toolkit/workspace/precompiled/web/filesui.html
toolkit/workspace/precompiled/web/filesui.js
```

fmandl / 27 file entries

```
micrOS/packages/alarm_system/package/LM_alarm_system.py
micrOS/packages/alarm_system/package/__init__.py
micrOS/packages/alarm_system/package/alarm_system/__init__.py
micrOS/packages/alarm_system/package/alarm_system/actions.py
micrOS/packages/alarm_system/package/alarm_system/door_sensor.py
micrOS/packages/alarm_system/package/alarm_system/event_log.py
micrOS/packages/alarm_system/package/alarm_system/mqtt_watcher.py
micrOS/packages/alarm_system/package/alarm_system/sms_handler.py
micrOS/packages/alarm_system/package/pacman.json
micrOS/packages/garage_remote/package/LM_garage.py
micrOS/packages/garage_remote/package/__init__.py
micrOS/packages/garage_remote/package/pacman.json
micrOS/packages/phone_manager/package/LM_users.py
micrOS/packages/phone_manager/package/__init__.py
micrOS/packages/phone_manager/package/manager.py
micrOS/packages/phone_manager/package/pacman.json
micrOS/packages/sim800/package/LM_sim800.py
micrOS/packages/sim800/package/LM_sim800_http.py
micrOS/packages/sim800/package/LM_sim800mqtt.py
micrOS/packages/sim800/package/__init__.py
micrOS/packages/sim800/package/codec.py
micrOS/packages/sim800/package/modem.py
micrOS/packages/sim800/package/notify.py
micrOS/packages/sim800/package/pacman.json
micrOS/source/LM_L298N.py
micrOS/source/LM_mqtt_client.py
micrOS/source/modules/LM_mqtt_client.py
```

Contributors' file changes

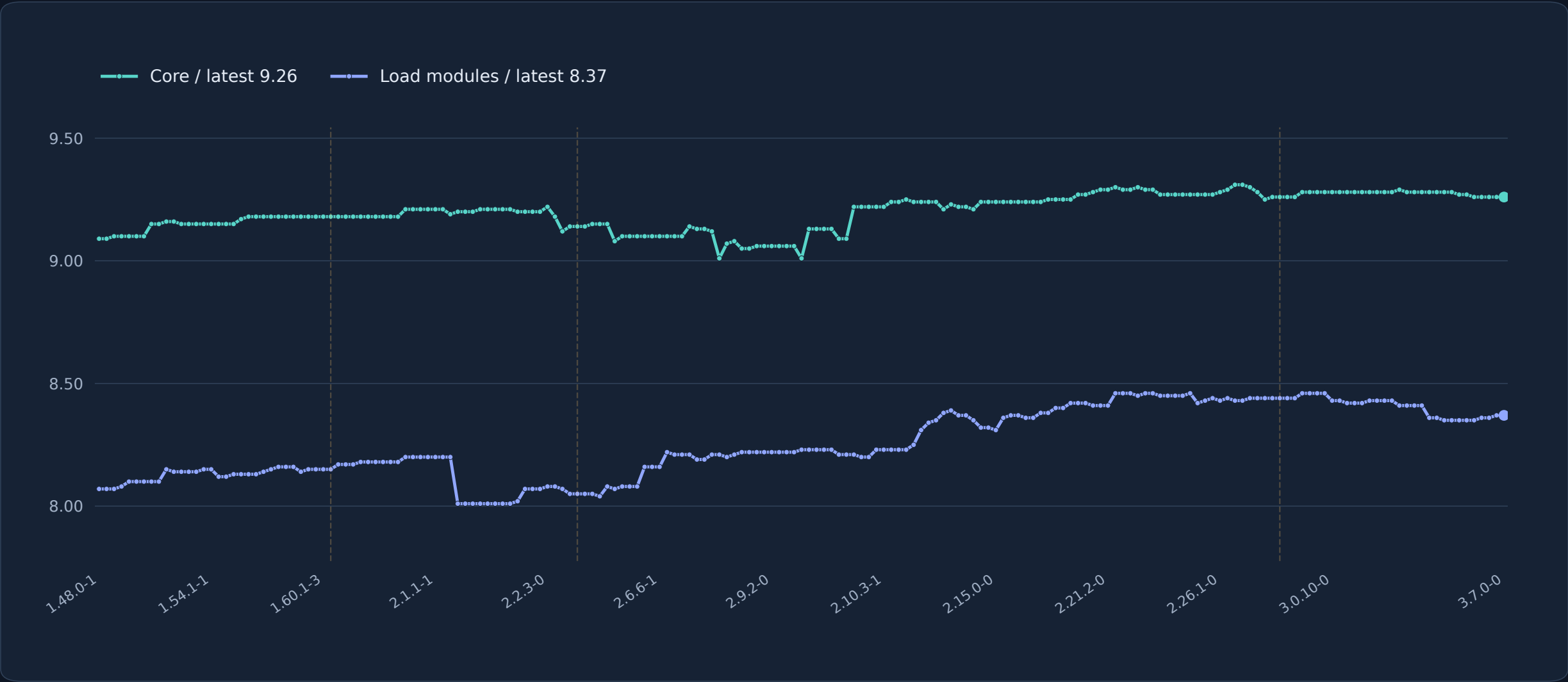
Complete recorded file lists / long paths wrap and continue onto the next card

davidandrasi / 3 file entries

tools/MicrOSDevEnv/MicrOSDevEnv.py
tools/MicrOSDevEnv/SearchDevices.py
tools/requirements.txt

Code quality

Pylint scores across every recorded version



Dependency warnings

Load module dependency checks across every recorded version

5 latest warnings



Version commit history

Latest 60 entries / newest first / full commit messages

Version	Commit ID	Message
3.7.0-0	9fc7afed3e994e6d8381c932bd447444454e7e3a	[micrOS][Core] Guard imported LM commands - version: 3.7.0-0
3.6.4-0	645d6f17d7c14782523e83d6d3e620b41f1d37d4	[micrOS][Core][Web] Persist web app settings - version: 3.6.4-0
3.6.3-1	c0b27f7988570fa27ce3a8514bfe49172df236d0	[micrOS][Web Config] Fix config UI drafts and Scheduler layout - 3.6.3-1
3.6.3-0	a903b75c81faffe79f9a96c8d83426ae5ef3b58b	[micrOS][WebConfig] Lazy-load Config Packages UI and mark installed packages
3.6.2-0	0699b389b7a5709a098cb3237b19daa48006cce5	micrOS/packages update
3.6.1-0	211417389d067b9826f8d950ed50f50d4ff418d8	[micrOS][webUI] Config/Scheduler updated configuration cards - version: 3.6.1-0
3.6.0-0	2fa75484b640a7d716240aaffe27abb102f09f	[micrOS][Core][PYPI] Add memory-aware WebUI defaults - version: 3.6.0-0
3.5.0-0	ca8f0dc0062bb4bbe301f283df3d5c83d369d69e	[micrOS][Web UI - Files] Improve fileserver UI and editor workflow
3.4.2-0	e6d22b9dc02053e1251a7ba4799207facaddbf99	[micrOS][Core] Harden web routing and frontend authentication - 3.4.2-0
3.4.1-0	2908010f3a267279d09fb75a3cf0f229b5da3cac	[micrOS][Web] Landing page web UI redesign - version: 3.4.1-0
3.4.0-1	c82376cae0663042a6107c386ddc0781c2e942da	Move LM_OV2640.py, LM_Id2410.py, LM_roboarm.py into packages + fix devToolkit --lint to include packages in core ref counts too
3.4.0-0	ccab5bed1904a5f631ba5e689e71b04b010d7af2	Config grouping visual update
3.3.2-0	3090979bdc9e1735c4831860c496d85a0e9e9de1	[micrOS][Core][Web] Return stream outputs directly and add Debug diagnostics UI - version: 3.3.2-0
3.3.1-0	d464b8e2c5544689848cdd15b849eea746dc3469	Minimal web config UI update (Pin-mapping -> Pinmap), and package update (irrigation_system)
3.3.0-0	f0a1e38df86b53b560841deed51f6262d33d2006	[micrOS][LoadModule] Add protected config reboot endpoint

Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
3.2.5-0	6cb0490381d7b22d79537b178acebb9751ed379b	[micrOS][PYPI] Web Config editor updates - version: 3.2.5-0
3.2.4-0	cc496a2a5b29ae02381d0ce46c374ad95dd056ee	[micrOS][WebConfig] Polish web UI controls and mobile REST input - 3.2.4-0
3.2.3-0	10336d15c21c71dd42ab4d14bf9ad9c443da4c18	[micrOS][WebConfig] Normalize scheduler and REST config UI - version: 3.2.3-0
3.2.2-0	5a0646f33a7cbc01a11a9d031ba93ebaa5c77979	[micrOS] Improve config web UI controls - version: 3.2.2-0
3.2.1-0	4c7464edeb2aadee61716934c8b5e8a8174d94bf	[micrOS][LoadModule] Improve fileserver directory metadata and mobile UI - version: 3.2.1-0
3.2.0-0	5c5c8c486e7b84b371cd8c96b3e1cf8479fb7bab	Config WebUI - sfuncman_3.2.0-0.json - version: 3.2.0-0
3.1.1-0	ca733296be5cc20fdff08c835d3cfea588d581ca	[micrOS] move LM_co2.mpy and LM_mh_z19c.py into co2 package, remove them from default LMs + package submodule update
3.1.0-0	79209de34dc7f59f7e607ab50b62597de6282d91	[devToolKit][ToolKit] Optimize web resources during cross compile + event based dashboard widgets - version: 3.1.0-0
3.0.10-0	49ff95b16ec2c367bbc864f883c9d64ef7a45282	Refactor modules into (installable) packages: LM_VL53L0X.py, LM_sound_event.py, LM_tcs3472.py
3.0.9-0	1204dfe4f925a3635401a0ef6afd9c73ce4142cb	[micrOS] Minor fixes in Files.py and LM_cluster.py, LM_pacman.py improvement (pack_ls command)
3.0.8-0	17916d676e005bb50ff9dac7ee873979af8a1f9d	[micrOS] Add uapi_widgets.js (modules helper auto load) for autocomplete
3.0.7-0	f7d45a039056ac2aafe279c5b5086e22e899398d	[micrOS][PYPI] web: split optional UI plugins and improve editor UX - version: 3.0.7-0
3.0.6-0	87d69a40f364f46040296dae1ba4be372e1efaaf	[micrOS] web config ui beta endpoints (/config (GET/POST), /config/ui (GET)) in LM_web.py + config.html template
3.0.5-0	d08dcc36f605d070001330e356f317b4220b196d	[micrOS] Web dashboard optimization and LM_OV2640 cam module integration

Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
3.0.4-0	f68555d913254d05b5b3130f17f156c2d0584a07	[microOS][PYPI] Config.py (read/write) improvements + utests/test_config.py + faster test_scheduler.py - version: 3.0.4-0
3.0.0-0	6faf9bb256bf882156f084f65e8caac11a8ef095	[microOS] Micro-improvements - string literal reduction
2.30.0-0	9a82cf056dc2695a5202de031ab8c117860db741	[microOS] HA and analysis updates - version: 2.30.0-0
2.29.0-0	7d0b7f3d882d160d73e83fb1eab0ed52efff673c	[microOS] Performance and stability improvements
2.28.0-0	82de41362c421f13b65e4d98bb8feffdd2d68ca5	[microOS] Boot sequence flow memory optimization and WDT integration into microOS task.feed (30 sec) - initialized from microOS.py - version: 2.28.0-0
2.27.2-0	3e07063afb6cdf5321093ae274a35f2eeb8aa81	Doc update microOS/ARCHITECTURE.md + APPLICATION_GUIDE.md -> microOS/MODULE_GUIDE.md
2.27.1-0	9a1fe1fa1a9c46b0c6033ecc7ca2bd5c87ce8a5f	[microOS] Fix unsupported media types issue in fs/editor for py, dat, cache, log types -> fallback to plain/text as txt - version: 2.27.1-0
2.27.0-0	9f9810051bd1e05d0e907786af6bc1bc44777e4c	[microOS] Fix simulator usecases (fs-editor is broken...) - version: 2.27.0-0
2.26.3-0	b68437c641d48f509f73cdb66814a3598e17a857	[microOS] Web UI enhancements (ustyle.css) - version: 2.26.3-0
2.26.1-0	bdab6d5927c16f2735e910aca22ca9b66307fb76	[microOS][PYPI] Minor improvements - LM_neomatrix.py migration under microSPackages - minor tldr improvements - release candidate (3.0) - version: 2.26.1-0
2.26.0-0	8ae5ee3ecc24a4ea7b061f2cf786ffe4dd41b1dd	[microOS] Small fix for LM_OV2640.py.photo() endpoint
2.25.3-0	247a27860eb8002edc0eb4076707c83e7cfafaae	[microOS][PYPI] 3.0.0 Candidate - Pacman.py (LM_pacman.py) package upgrade feature version: 2.25.3-0
2.25.2-1	c8e0df15a4c9a43ce8de2968ee72607cbf2993b6	LM_OV2640.py photo endpoint - filesaver improvement: can take multiple photos with timestamp up to storage 60% - filserver usage callback reuse in OV2640 (PoC)

Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
2.25.2-0	30d2651194defeb1364a2bf1bc85aaaa49e68be2	[micrOS] Web Homepage (index.html) visual improvements: endpoint groups :)
2.25.1-0	bdad9ac20f27b3f44d290983ec1f8c383376e053	[micrOS] Web.py generic plain/text fallback (only check /web root file extensions) + OV2640 fileserver integration with /cam/photo endpoint
2.25.0-0	f24b37fa53b1835d1693f19195967a61bcb06cec	[micrOS][PYPI] Web.py->web_endpoints enhancement - version: 2.25.0-0
2.24.1-0	0c0b66499f0376795be4680c842d56bcabfa1350	[micrOS] simulator_lib/uasyncio.py wait_for() fix + micrOS Files WebUI beta and fileserver enhancements - version: 2.24.1-0
2.24.0-0	611f1757de64201305699c13b904acc75bff098e	Doc update and updated precomiled files
2.23.1-0	e7e6ba28fd51be3d5ee96b17a25e5ad5a3fe87d5	[micrOS] Espnow.py less strict remote command execution (restrct unloaded modeules execution with auth:True in config) + Fix dashboard UI node status visualization - version: 2.23.1-0
2.23.0-0	4c29abe36f2933f476f6845d252bbaf9852ba9b7	[micrOS][PYPI] Fix USB configuration issues (backup/restore) + remove legacy telegram features, from now telegram is only a LoadModule with micrOS system Notify integration - version: 2.23.0-0
2.22.1-0	718fc75f7f21ecd7588ed449f81b5368481a12f8	[micrOS] InterConnect, Espnow improvement - version: 2.21.3-0
2.22.0-0	014d55d7a2ee120a2031656d0694cf20e8bd8c8a	[devToolKit] toolkit/socketClient.py embedding fix into devToolKit.py -c with -p param fixes
2.21.4-0	103166148a01a6057b8ffd65ef74ccac7d8edb07	[micrOS] Pacman.py.install local mip registry unpack support quick fix and micrOS/packages update
2.21.3-0	6844ebcdf42a1902a67a93d7bd43711195f86fdd	[DevToolKit] New micrOS/packages --unpack handles 3PP packages, toolkit/simulator_lib/simulator.py: Monkey-patch time module duw to 3PP dependencies, remove hardcoded toolkit/simulator_lib/mqtt_as dependency
2.21.2-0	0db0f84878ce0096bf9056cd5b5a21de4e4145f8	[micrOS] microlO.py and LM_system.py pinmap update, show avaialble pinmap names as well

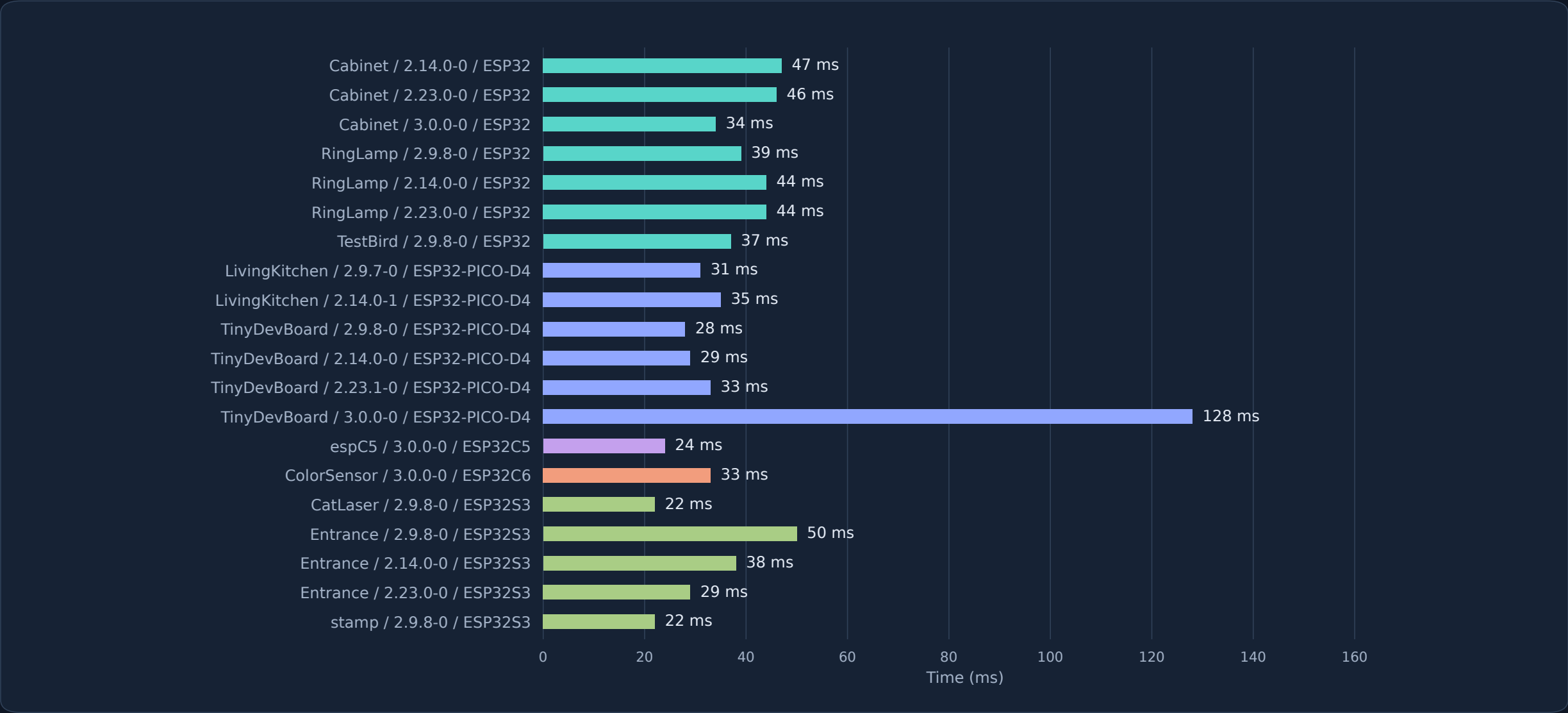
Version commit history

Latest 60 entries / newest first / full commit messages / continued

Version	Commit ID	Message
2.21.1-0	cb3ce11a779c718f1466b8ca73d511412040df4d	[microOS] WebEngine refactor with new dimensioning solution (is_limited and chunking size adjustments): >20kb No webUI, 20-100kb use default chunking settings, <100kb upscale chunking size to enhance file tansaction speed - version: 2.21.1-0
2.21.0-0	31e5b9a34100415cbf5ba2053d0ae273990b0e70	[microOS] REPL helper/tools: helpers.py: reset and shell shortcuts, shell makes possible to run ShellCli interface over UART (REPL) (same as socket ShellCli) BETA
2.20.2-0	d33cc344de4ec0703db7a733ef86411d180fdbd1	ESPNow: module name added to espnow task ID
2.20.1-0	484d1d813649413512b75c1a2d641377e622a307	ESPNow: transaction ID implementation and refactor
2.20.0-0	dd5060453b49c41f0579fc43c6337526dbc45e65	[microOS] Unified task (micro_task) creation response: {'task_tag': 'verdict'} - version: 2.20.0-0
2.19.0-0	ed7aef655490199721f9f65be19746c8ea312ad4	[microOS][PYPI] InterConnect.py logging fix, socketClient.node_status feature status addition - version: 2.19.0-0

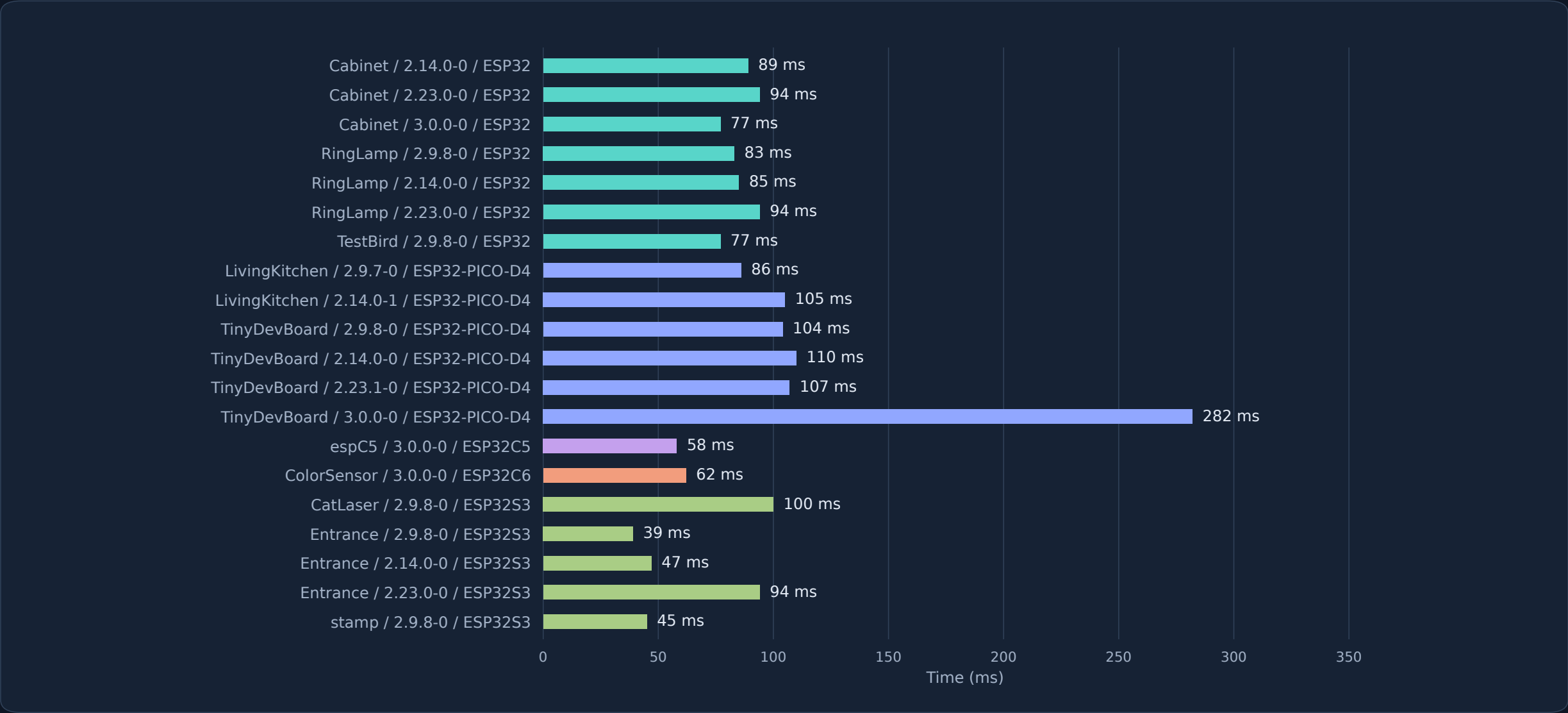
Shell heartbeatx10 response

Device benchmarks / records 1-20 of 20



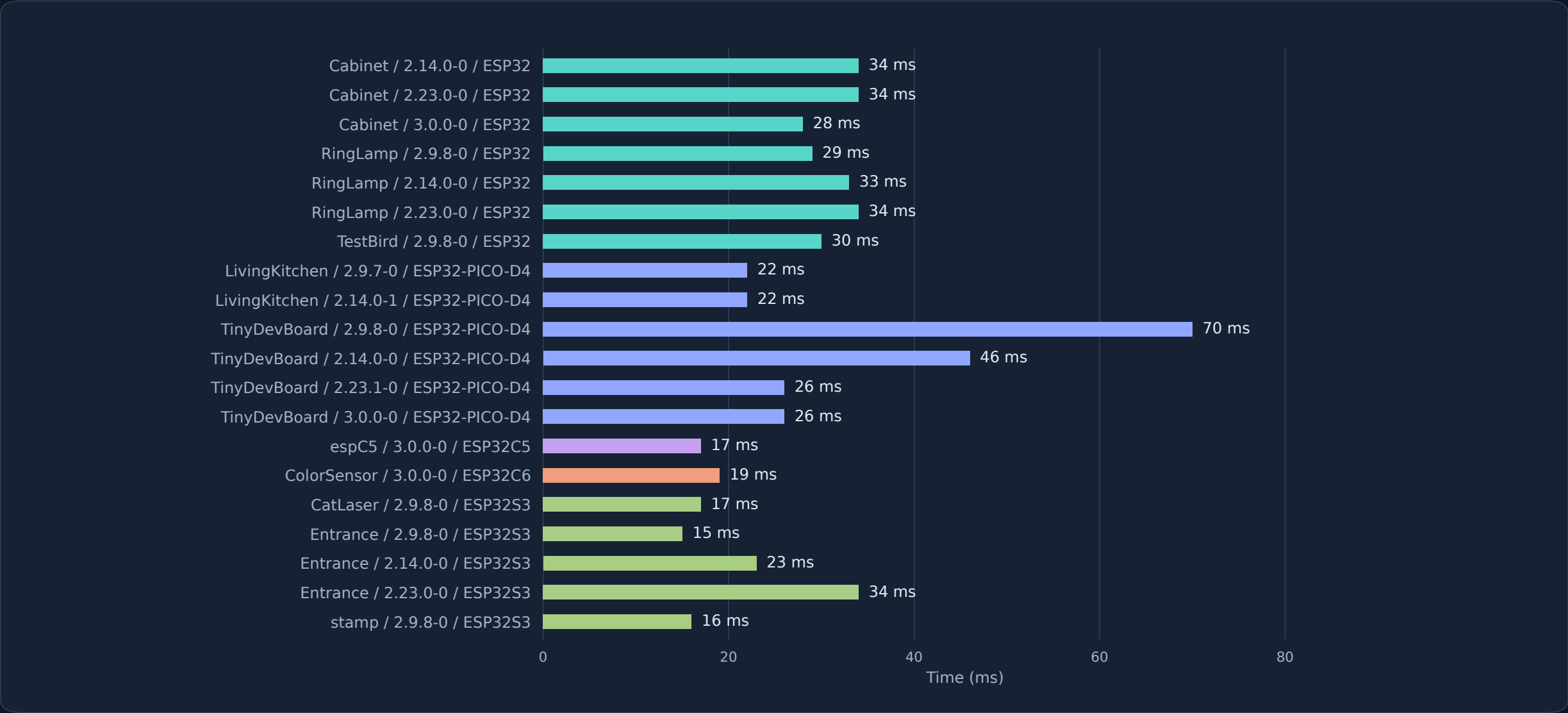
Shell multi session dt

Device benchmarks / records 1-20 of 20



Shell single session dt

Device benchmarks / records 1-20 of 20



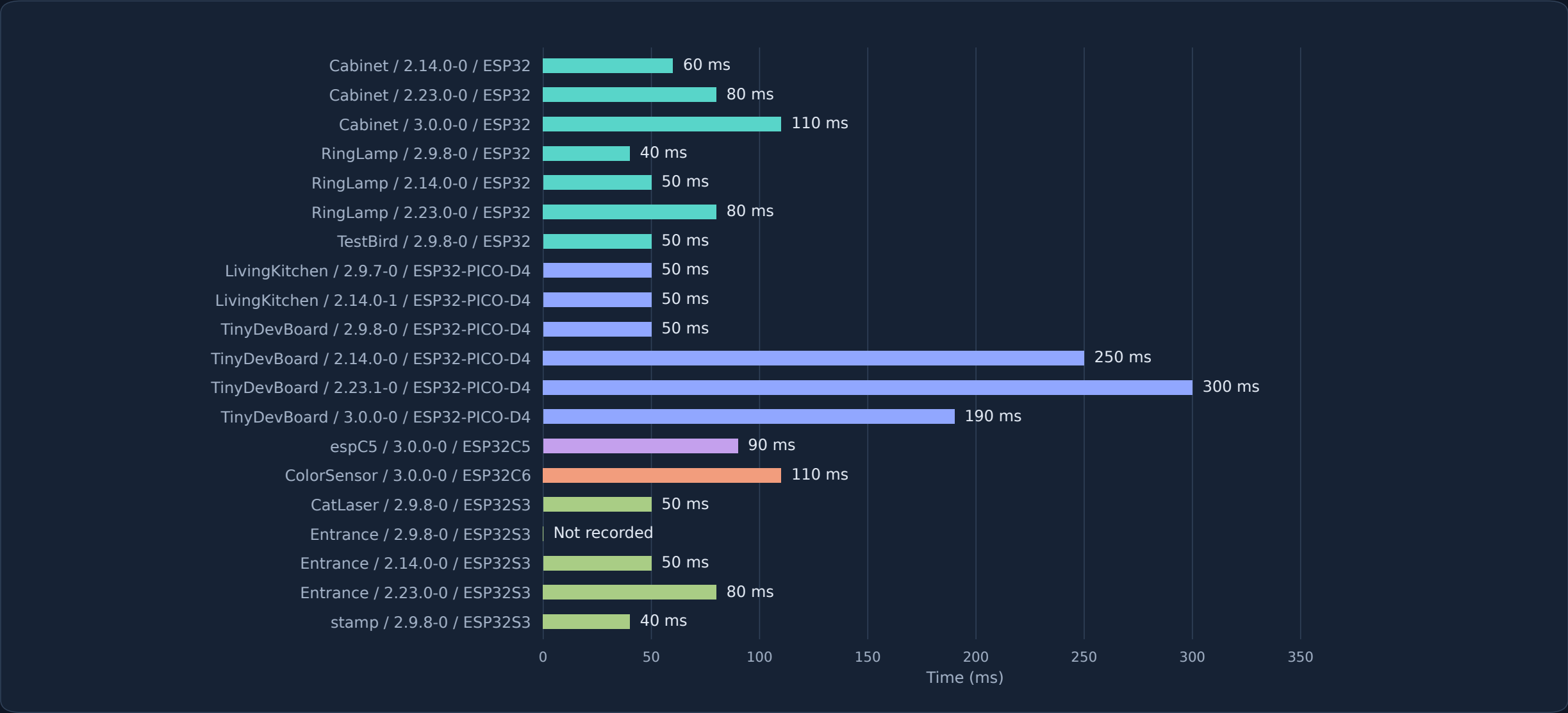
Web landingpage response

Device benchmarks / records 1-20 of 20



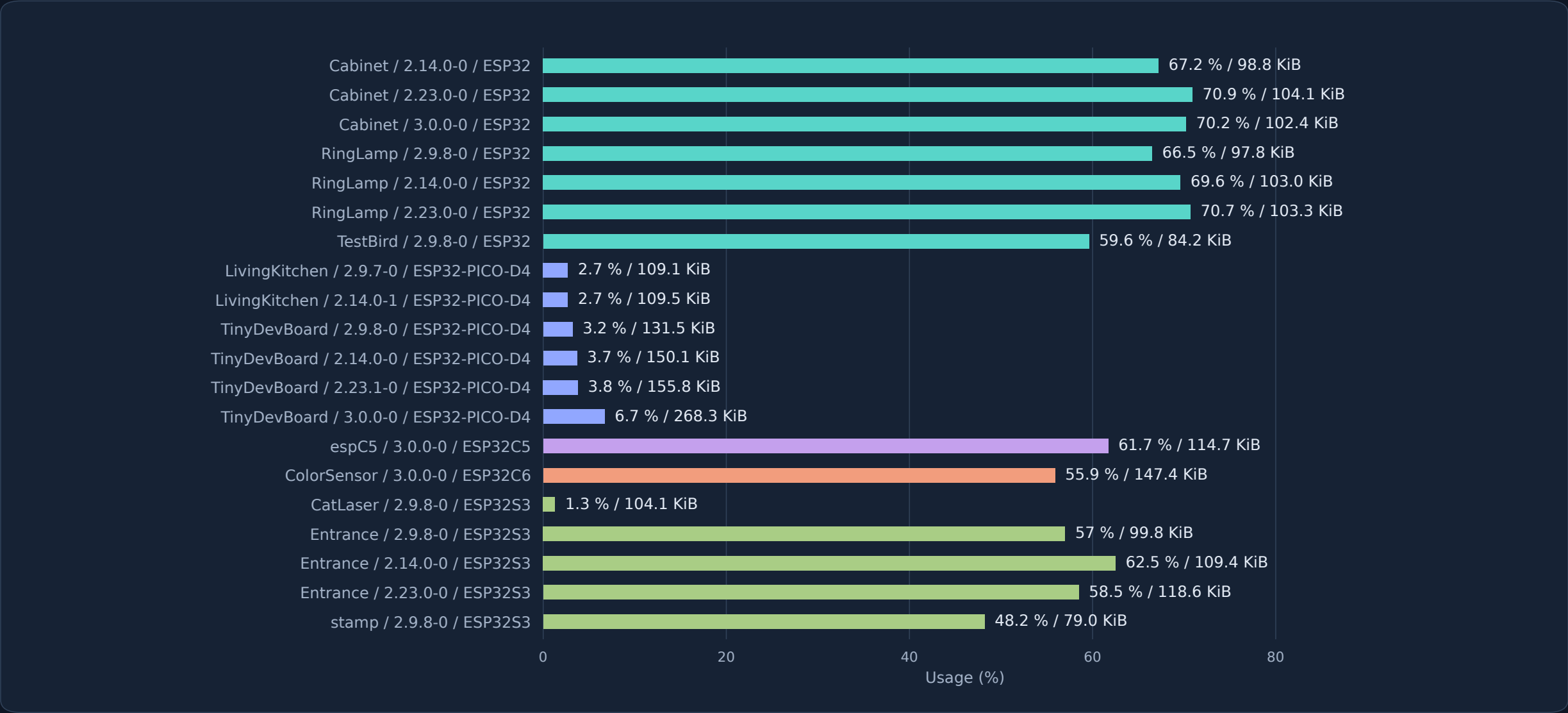
Web rest response

Device benchmarks / records 1-20 of 20



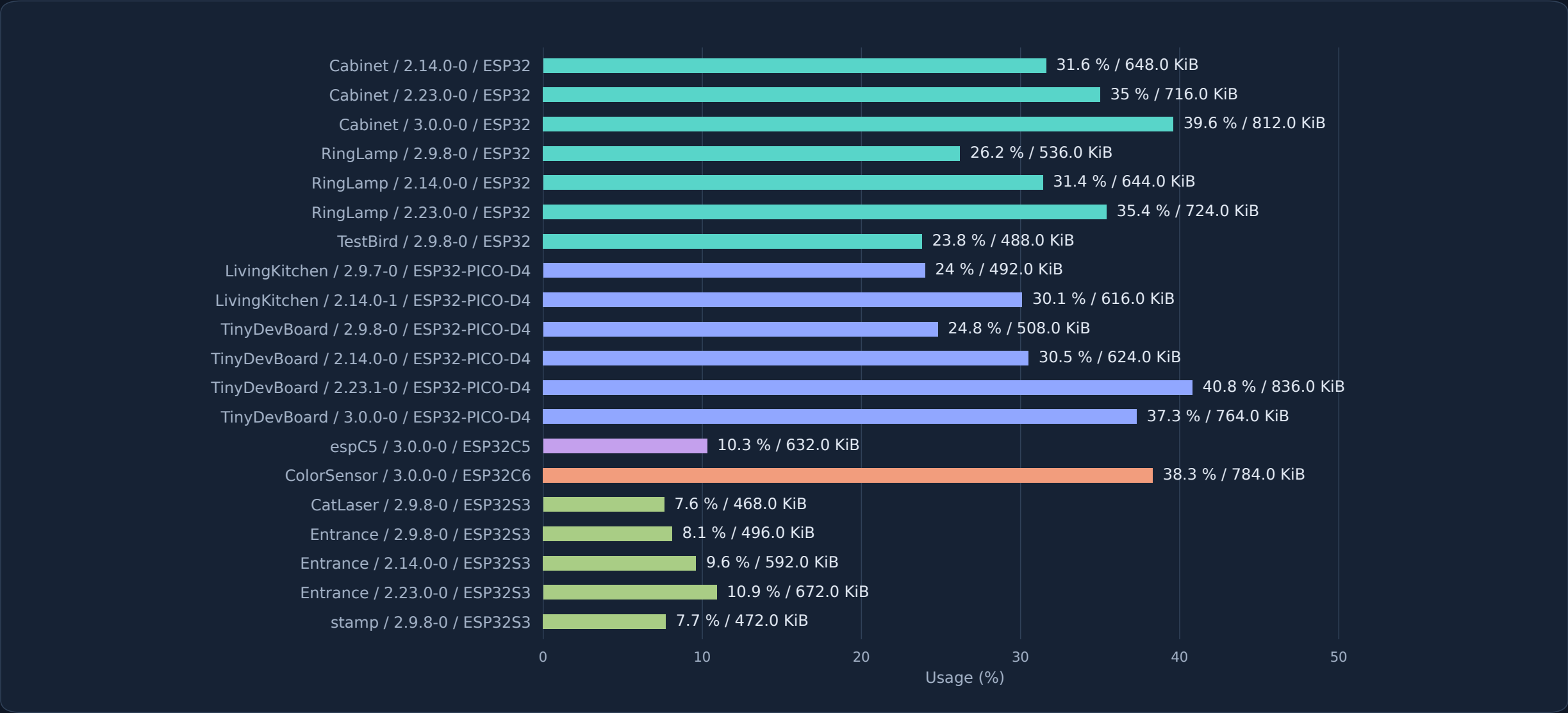
Memory utilization

Device benchmarks / records 1-20 of 20



Filesystem utilization

Device benchmarks / records 1-20 of 20



Device module inventory

Complete loaded-module lists for each benchmark record

Device / version	Device type	Loaded modules
Cabinet / 2.14.0-0	ESP32	system, cct, dht22, task
Cabinet / 2.23.0-0	ESP32	system, cct, dht22, task
Cabinet / 3.0.0-0	ESP32	system, cct, dht22, task
RingLamp / 2.9.8-0	ESP32	neopixel, pacman, system, task
RingLamp / 2.14.0-0	ESP32	neopixel, pacman, system, task
RingLamp / 2.23.0-0	ESP32	neopixel, system, task
TestBird / 2.9.8-0	ESP32	types_demo, system, task
LivingKitchen / 2.9.7-0	ESP32-PICO-D4	esp32, rgbcct, dimmer, rgb, cct, dashboard_be, system, intercon, task
LivingKitchen / 2.14.0-1	ESP32-PICO-D4	rgbcct, dimmer, rgb, cct, dashboard_be, system, task
TinyDevBoard / 2.9.8-0	ESP32-PICO-D4	system, rgb, telegram, cct, dashboard_be, intercon, oled_ui, oled, gameOfLife, task
TinyDevBoard / 2.14.0-0	ESP32-PICO-D4	system, rgb, pacman, cct, telegram, dashboard_be, oled, oled_ui, gameOfLife, task
TinyDevBoard / 2.23.1-0	ESP32-PICO-D4	system, rgb, telegram, cct, pacman, cluster, dashboard_be, oled, oled_ui, gameOfLife, task
TinyDevBoard / 3.0.0-0	ESP32-PICO-D4	gameOfLife, oled_ui, web, fileserver, dimmer, telegram, system, rgb, cluster, cct, oled, task
espC5 / 3.0.0-0	ESP32C5	system, task
ColorSensor / 3.0.0-0	ESP32C6	web, tcs3472, system, cluster, neopixel, fileserver, task

Device module inventory

Complete loaded-module lists for each benchmark record / continued

Device / version	Device type	Loaded modules
CatLaser / 2.9.8-0	ESP32S3	roboarm, telegram, system, dashboard_be, servo, switch, task
Entrance / 2.9.8-0	ESP32S3	telegram, system, buzzer, pacman, task
Entrance / 2.14.0-0	ESP32S3	pacman, buzzer, system, telegram, task
Entrance / 2.23.0-0	ESP32S3	buzzer, telegram, system, pacman, task
stamp / 2.9.8-0	ESP32S3	system, task